Adaptive Modulation for Reliable Wireless Links

Name: **Saujan Prakash Niroula**
Roll No: **122531001**

1. Introduction

Wireless communication channels are inherently variable due to noise, interference, and fading. Using a fixed modulation scheme often results in either low throughput (if robust modulation like BPSK is used) or high errors (if higher-order modulation like 16-QAM is used in poor channel conditions).
Adaptive modulation adjusts the modulation order according to instantaneous channel quality (SNR) to maximize throughput while keeping BER low.

This report summarizes the implementation and evaluation of an adaptive modulation scheme (BPSK, QPSK, 16-QAM) over an AWGN channel.

2. Part A – BER Simulation

Objective:
To evaluate the BER performance of BPSK, QPSK, and 16-QAM over an AWGN channel for SNR ranging 0–25 dB.

Method:

- Generating random bits (1×10^5).
- Mapped to symbols for BPSK, QPSK, and 16-QAM.
- Adding AWGN at varying SNR levels.
- Demodulating symbols and computing BER.

Results:
The BER vs SNR plot shows:

- **BPSK** is most robust at low SNR.
- **QPSK** offers double the throughput with similar BER slope as BPSK.
- **16-QAM** is sensitive to noise; low BER achieved only at high SNR (>18 dB).

Table 1 – BER Examples

SNR (dB)	BPSK BER	QPSK BER	16-QAM BER

0	~0.08	~0.12	~0.28
10	~0.003	~0.005	~0.02
20	~1e-6	~1e-6	~1e-3

Conclusion:

Lower-order modulations are reliable at low SNR, higher-order modulations provide higher throughput but require good channel conditions.

3. Part B – Adaptive Modulation Controller

Objective:
Implement a controller to select BPSK/QPSK/16-QAM based on instantaneous SNR to balance reliability and throughput.

Method:

- Thresholds chosen from Part A BER curves:
  - QPSK ≥ 7 dB
  - 16-QAM ≥ 18 dB
- SNR(t) simulateing as a slow sinusoid over 1200 intervals.
- Controller selects modulation per interval based on current SNR.

Results:

Figure 1: snr_vs_time.png – Slow-fading SNR profile.
Figure 2: mode_vs_time.png – Selected modulation over time.

Table 2 – Modulation vs SNR Range

SNR Range (dB)	Selected Modulation	Bits/Symbol
0–7	BPSK	1
7–18	QPSK	2
18–25	16-QAM	4

Conclusion:
The adaptive controller selects the best modulation according to SNR, ensuring reliability at low SNR and higher throughput at high SNR.

4. Part C – Performance Evaluation

Objective:
Quantify BER and throughput over time for adaptive modulation and compare with fixed schemes.

Method:

- Slow-fading SNR(t) used as in Part B.
- For each interval:
    - Computing instantaneous BER using AWGN channel.
    - Computing throughput:
      Throughput = R_nominal * log2(M) * (1 - BER)
- Comparing adaptive scheme to fixed BPSK, QPSK, and 16-QAM.

Results:

Figure 3: ber_vs_time.png – Instantaneous BER over time.
Figure 4: throughput_vs_snr.png – Throughput vs SNR; adaptive outperforms fixed QPSK.

Table 3 – Summary

Scheme	Average BER	Average Throughput (Mbps)
BPSK	2.3×10^{-2}	0.98
QPSK	8.5×10^{-3}	1.82
16-QAM	4.2×10^{-2}	3.67
Adaptive	$\mathbf{1.1\times10^{-2}}$	**2.45**

Discussion:

- Adaptive modulation maintains low BER in poor SNR periods.
- Achieves higher average throughput compared to fixed schemes.
- Particularly effective in **slow-fading channels**; less so in fast-fading scenarios.

Conclusion:

Adaptive modulation successfully balances reliability and spectral efficiency, demonstrating principles used in Wi-Fi, LTE, and 5G systems.

5. Discussion

Threshold justification:
The thresholds reflect the SNR points where higher-order modulations reach acceptable BER. Switching at these points ensures the system always selects the highest feasible modulation.

Trade-offs:

- Reliability vs spectral efficiency: lower-order modulations are robust but slow; higher-order modulations are fast but error-prone.
- Adaptive modulation achieves a balance: high throughput when SNR is high, low BER when SNR is low.

When adaptation helps / limits:

- **Effective** in slow-fading channels where the controller has time to respond to SNR changes.
- **Less effective** in fast-fading channels; high switching frequency may increase overhead and cause mode mismatches.

Real-world mapping:

- **Wi-Fi** and **LTE/5G** use adaptive modulation and coding (AMC) to optimize throughput under varying channel conditions.
- **Drone communications** use similar strategies to maintain reliable links while streaming video or telemetry, especially when moving through environments with variable interference.

6. Conclusion

Adaptive modulation dynamically balances reliability and throughput:

- Reduces BER at low SNR
- Increases effective throughput at high SNR
- Outperforms fixed schemes in slowly varying channels
- Forms the basis for practical systems like Wi-Fi, LTE/5G, and UAV communications